

## PEMBRY SANDS, United Kingdom

WMO: 036050

Lat: 51.7144N

Lon: 4.3675W

Elev: 22

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | 26.1       | 28.6 | 21.2                          | 15.9 | 30.6 | 24.0 | 18.2 | 31.9 | 49.1                  | 47.7 | 42.3 | 47.6 | 5.3                      | 70   | 0.727 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.9                   | 73.7            | 64.5 | 70.7 | 62.9 | 68.2 | 61.8 | 66.0                | 71.2 | 64.4 | 68.1 | 63.2 | 66.5 | 10.4                 | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.1                            | 89.7 | 67.2 | 62.9 | 86.0 | 65.8 | 61.9 | 82.8 | 64.8 | 30.7          | 71.4 | 29.5 | 68.3 | 28.7 | 66.6 | 81.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 39.8              | 34.1 | 30.2 | DB                         | 20.7 | 80.6               | 3.2 | 4.0                                                | 18.4 | 83.5       | 16.5 | 85.9       | 14.7 | 88.2       | 12.4 | 91.1 |
|                   |      |      | WB                         | 19.8 | 69.9               | 3.4 | 3.5                                                | 17.4 | 72.5       | 15.4 | 74.5       | 13.6 | 76.5       | 11.1 | 79.0 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 50.8   | 42.2  | 42.1  | 44.2  | 47.8  | 52.7  | 57.5  | 60.9  | 60.7  | 57.5  | 52.9  | 47.1  | 43.3 |
|                                                                                  | DBStd       | 8.41   | 5.55  | 4.98  | 4.68  | 4.63  | 4.52  | 3.96  | 3.64  | 3.63  | 4.06  | 5.22  | 5.74  | 6.41 |
|                                                                                  | HDD50       | 1133   | 244   | 223   | 183   | 91    | 22    | 1     | 0     | 0     | 2     | 33    | 119   | 215  |
|                                                                                  | HDD65       | 5209   | 708   | 642   | 644   | 515   | 382   | 228   | 137   | 140   | 227   | 376   | 538   | 672  |
|                                                                                  | CDD50       | 1420   | 1     | 0     | 4     | 26    | 106   | 226   | 337   | 332   | 227   | 122   | 31    | 8    |
|                                                                                  | CDD65       | 21     | 0     | 0     | 0     | 0     | 1     | 3     | 9     | 6     | 2     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 102    | 0     | 0     | 0     | 0     | 4     | 19    | 48    | 29    | 2     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 13     | 0     | 0     | 0     | 0     | 0     | 1     | 7     | 5     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 13.8   | 15.2  | 14.7  | 13.3  | 12.5  | 13.0  | 12.7  | 12.7  | 13.1  | 13.3  | 14.9  | 14.6  | 15.2 |
| Precipitation                                                                    | PrecAvg     | 47.9   | 5.0   | 3.7   | 3.2   | 2.8   | 3.0   | 3.1   | 3.2   | 4.1   | 3.5   | 5.6   | 5.6   | 5.2  |
|                                                                                  | PrecMax     | 57.6   | 8.8   | 7.3   | 6.4   | 5.5   | 6.1   | 7.4   | 8.6   | 8.0   | 6.9   | 9.3   | 10.4  | 9.0  |
|                                                                                  | PrecMin     | 38.1   | 0.6   | 1.2   | 1.5   | 0.8   | 0.7   | 1.1   | 0.6   | 0.8   | 1.2   | 1.7   | 2.3   | 1.8  |
|                                                                                  | PrecStd     | 5.6    | 2.0   | 1.8   | 1.3   | 1.4   | 1.3   | 1.6   | 1.9   | 1.8   | 1.6   | 2.1   | 2.0   | 2.0  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 52.7  | 54.0  | 61.5  | 67.3  | 72.9  | 76.5  | 79.3  | 78.3  | 73.0  | 65.7  | 59.0  | 55.4 |
|                                                                                  |             | MCWB   | 50.8  | 48.7  | 52.3  | 56.2  | 60.9  | 65.0  | 66.5  | 68.9  | 65.0  | 60.2  | 56.3  | 53.6 |
|                                                                                  | 2%          | DB     | 51.4  | 50.8  | 55.3  | 61.2  | 67.4  | 71.1  | 73.6  | 72.1  | 68.6  | 62.8  | 57.3  | 54.0 |
|                                                                                  |             | MCWB   | 49.9  | 48.3  | 49.8  | 53.3  | 58.8  | 62.7  | 64.5  | 64.5  | 62.0  | 59.1  | 55.5  | 52.4 |
|                                                                                  | 5%          | DB     | 50.3  | 49.7  | 52.4  | 57.5  | 63.3  | 67.6  | 70.4  | 68.8  | 66.2  | 61.0  | 56.1  | 53.0 |
|                                                                                  |             | MCWB   | 48.9  | 47.7  | 48.4  | 51.2  | 56.5  | 60.8  | 63.3  | 62.4  | 60.5  | 58.4  | 54.2  | 51.4 |
|                                                                                  | 10%         | DB     | 49.4  | 48.7  | 50.6  | 54.9  | 60.4  | 64.8  | 67.8  | 66.8  | 64.3  | 59.9  | 54.9  | 51.8 |
|                                                                                  |             | MCWB   | 47.9  | 46.9  | 47.7  | 50.3  | 55.2  | 59.3  | 62.1  | 61.4  | 60.1  | 57.4  | 52.9  | 50.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 51.8  | 50.7  | 53.7  | 57.5  | 63.2  | 67.2  | 68.1  | 69.7  | 65.9  | 61.8  | 57.5  | 54.3 |
|                                                                                  |             | MCDB   | 52.2  | 51.6  | 58.7  | 65.4  | 70.0  | 73.6  | 75.4  | 76.4  | 70.2  | 64.2  | 57.8  | 54.7 |
|                                                                                  | 2%          | WB     | 50.4  | 49.4  | 51.0  | 54.3  | 59.9  | 63.7  | 65.9  | 65.6  | 63.8  | 60.4  | 56.0  | 53.1 |
|                                                                                  |             | MCDB   | 50.9  | 50.2  | 53.9  | 58.9  | 65.5  | 69.2  | 71.5  | 69.7  | 66.6  | 61.7  | 56.7  | 53.7 |
|                                                                                  | 5%          | WB     | 49.4  | 48.4  | 49.6  | 52.6  | 57.8  | 61.8  | 64.4  | 64.0  | 62.5  | 59.2  | 54.8  | 52.0 |
|                                                                                  |             | MCDB   | 50.1  | 49.2  | 51.3  | 55.8  | 61.9  | 66.3  | 68.5  | 66.6  | 64.7  | 60.4  | 55.8  | 52.7 |
|                                                                                  | 10%         | WB     | 48.4  | 47.5  | 48.6  | 51.2  | 56.0  | 60.2  | 63.0  | 63.0  | 61.2  | 58.0  | 53.4  | 50.7 |
|                                                                                  |             | MCDB   | 49.2  | 48.4  | 50.0  | 53.8  | 59.3  | 63.7  | 66.5  | 65.5  | 63.5  | 59.5  | 54.6  | 51.5 |
| Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 8.7   | 9.7   | 11.3  | 12.9  | 13.1  | 12.4  | 11.9  | 11.9  | 12.6  | 10.9  | 10.2  | 9.2  |
|                                                                                  | 5% DB       | MCDBR  | 7.0   | 9.2   | 14.0  | 17.2  | 18.8  | 19.3  | 18.0  | 15.8  | 15.8  | 9.8   | 7.9   | 6.8  |
|                                                                                  |             | MCWBR  | 7.4   | 7.9   | 10.2  | 11.1  | 12.0  | 12.3  | 11.1  | 10.2  | 10.9  | 7.7   | 7.6   | 7.3  |
|                                                                                  | 5% WB       | MCDBR  | 7.0   | 7.4   | 10.7  | 14.1  | 16.3  | 16.1  | 14.8  | 12.0  | 12.1  | 7.8   | 6.9   | 6.8  |
|                                                                                  |             | MCWBR  | 7.7   | 7.1   | 8.8   | 10.1  | 11.3  | 11.2  | 10.0  | 8.9   | 9.7   | 7.1   | 7.2   | 7.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.333  | 0.345 | 0.374 | 0.389 | 0.397 | 0.385 | 0.388 | 0.372 | 0.365 | 0.355 | 0.331 | 0.322 |      |
|                                                                                  | taud        | 2.331  | 2.373 | 2.313 | 2.298 | 2.319 | 2.380 | 2.377 | 2.438 | 2.447 | 2.433 | 2.386 | 2.308 |      |
|                                                                                  | Ebn at Noon | 210    | 244   | 259   | 269   | 272   | 276   | 273   | 272   | 260   | 238   | 210   | 191   |      |
|                                                                                  | Edh at Noon | 21     | 26    | 33    | 37    | 38    | 36    | 36    | 32    | 29    | 24    | 19    | 18    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 236    | 477   | 866   | 1377  | 1692  | 1839  | 1681  | 1402  | 1039  | 577   | 292   | 179   |      |
|                                                                                  | RadStd      | 23     | 44    | 80    | 104   | 105   | 160   | 142   | 96    | 76    | 47    | 24    | 22    |      |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (51 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page